RouteEd

Local Run Manual

Start the complete AI-powered personalized learning platform on your Windows computer. This guide covers services, the React interface, demo accounts, and the administrator-to-learner journey.

What starts	Where it runs	Purpose
React + Vite	http://localhost:5173	Web application and login
Spring Boot API	http://localhost:8080/api	Authentication, courses, analytics, AI routes
PostgreSQL	localhost:5432	Application data
Ollama	localhost:11434	Local llama3.2:3b personalization
Mailpit	http://localhost:8025	Safe local email inbox

Before you begin

Install Docker Desktop and Node.js 18 or newer. Docker runs PostgreSQL, Ollama, Mailpit, and the Spring Boot API; Java is therefore not required on the host for normal local startup. Keep Docker Desktop running before following the steps.

Project folder

```
C:\Users\GEETHABALU\OneDrive\Documents\ChatGPT\RouteEd
```

1. Start RouteEd services

Open PowerShell in the project folder and run the following command. On the first run Docker downloads images and builds the API, so allow a few minutes.

```
docker compose up -d --build
docker compose ps
```

Expected result

The output should show four running containers: **routed_api**, **routed_postgres**, **routed_ollama**, and **routed_mailpit**.

2. Ensure the local model exists

The personalization API uses the local **llama3.2:3b** model. Run this only if Ollama does not already show that model:

```
docker exec routed_ollama ollama pull llama3.2:3b
```

3. Start the frontend

Open a second PowerShell terminal, then run:

```
cd frontend
npm install
npm run dev -- --host 0.0.0.0
```

Open **http://localhost:5173** in your browser. You can verify the backend at **http://localhost:8080/api/health**; it should return status **UP**.

2. Sign in and orient yourself

Use a seeded development account so that the complete demo data is ready immediately.

Role	Email	Password	Use it for
Administrator	admin@routeed.demo	RouteEd@123	Admin Workspace and Course Builder
Learner	ava@routeed.demo	RouteEd@123	Python learner journey
Learner	ben@routeed.demo	RouteEd@123	Java learner journey and email
Learner	chitra@routeed.demo	RouteEd@123	Fresh learner test account

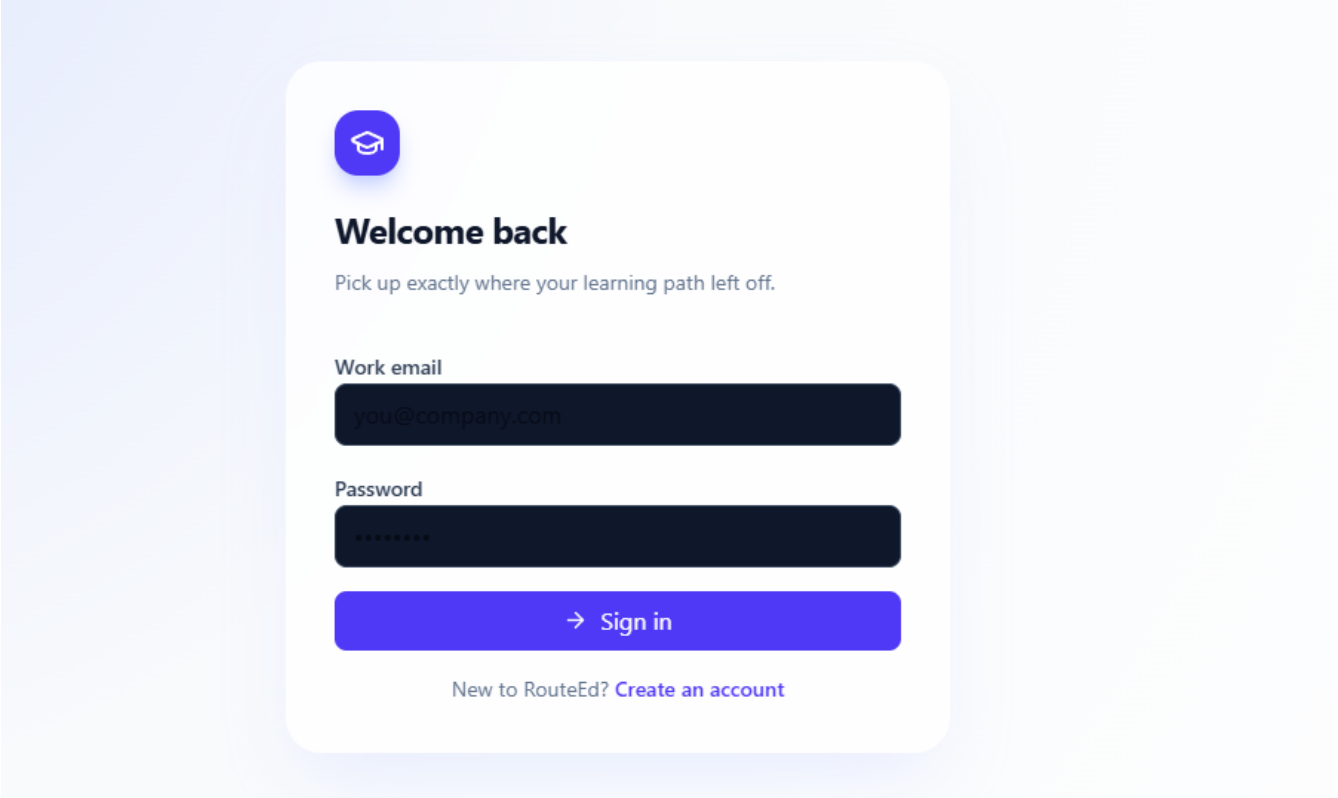


Figure 1. RouteEd sign-in screen. Enter one of the development account email/password pairs above, then select Sign in.

Important

These credentials are for local demonstration only. Public registration creates learner accounts; administrator accounts are protected from public sign-up.

3. Administrator workflow

Sign in with **admin@routeed.demo**, then use the left navigation.

Admin Workspace - http://localhost:5173/admin

Select Java Foundations, Python Foundations, or a course you created. The workspace provides learner-route count, completion rate, drop-off heatmap, and learner health scores. This is where a course administrator identifies learners who need attention.

Course Builder - http://localhost:5173/courses

Follow the three workflow cards in order:

Step	Action	Result
1. Course details	Enter a title, learner outcome, and completion days.	A new course is created and selected.
2. Learning modules	Add title, skill tag, optional content URL, description, and duration.	The ordered module list grows.
3. Diagnostic assessment	Add a question, one answer per line, the exact correct answer, and skill tag. Repeat for more questions.	Learners can take the diagnostic and receive a route.

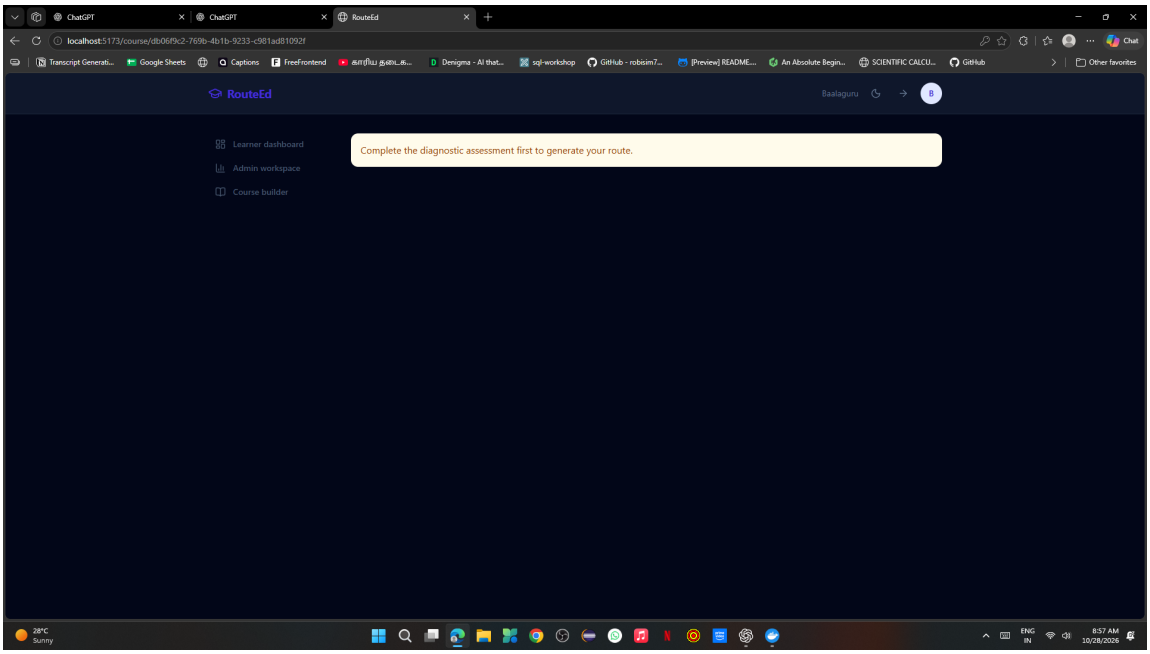


Figure 2. RouteEd learning workspace orientation. After an assessment is completed, the learner path becomes available from this area.

4. Learner and email workflow

Sign out, then sign in as a learner such as **ava@routeed.demo** or **chitra@routeed.demo**.

Take the diagnostic

1. Open the learner dashboard. 2. Select a course and choose Take assessment. 3. Answer all questions. 4. Select Finish assessment. RouteEd submits results to the API and local Ollama creates a personalized module order. Local models can take several seconds; keep the page open until the completion result appears.

Open the personalized route

Open the course route after the result screen. Mark modules complete as you work. Progress and the learner health score update for the administrator.

View email safely in Mailpit

A welcome email is generated when a route is created. Open **http://localhost:8025** to see messages without sending anything to a real inbox.

Stop the project

```
docker compose down
# In the frontend terminal, press Ctrl+C
```

Troubleshooting

Symptom	Check
Page does not open	Confirm the Vite terminal is still running and visit http://localhost:5173.
API or login error	Run docker compose ps, then visit http://localhost:8080/api/health.
Assessment takes longer	Ollama is local. Confirm routed_ollama is running and the llama3.2:3b model is installed.
No development email	Confirm routed_mailpit is running, then refresh http://localhost:8025.

You are ready to demonstrate RouteEd end to end: administrator creates content, learner completes a diagnostic, AI generates a learning path, progress is tracked, and the administrator reviews analytics.